## MISC

**Quadratic formula**:

$$ax^2 + bx + c = 0 \Rightarrow x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

**Monotonically in-/decreasing**:

$$\forall x.\ (f'(x) > 0) \Rightarrow a > b \Leftrightarrow f(a) > f(b)$$
$$\forall x.\ (f'(x) < 0) \Rightarrow a > b \Leftrightarrow f(a) < f(b)$$

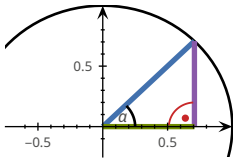
**RSS**:

$$\text{RSS} = \sum_i (y_i - f(x_i))^2$$

## TRIGONOMETRY

| $x$ | 0 | $30° = \pi/6$ | $45° = \pi/4$ | $60° = \pi/3$ | $90° = \pi/2$ |
|---|---|---|---|---|---|
| $\sin(x)$ | 0 | $1/2$ | $\sqrt{2}/2$ | $\sqrt{3}/2$ | 1 |
| $\cos(x)$ | 1 | $\sqrt{3}/2$ | $\sqrt{2}/2$ | $1/2$ | 0 |
| $\tan(x)$ | 0 | $1/\sqrt{3}$ | 1 | $\sqrt{3}$ | |

$$\tan(x) = \frac{\sin(x)}{\cos(x)}$$
$$\sin(2x) = 2\sin(x)\cos(x)$$
$$\sin^2(x) + \cos^2(x) = 1$$
$1$, $\sin(\alpha)$, $\cos(\alpha)$




## VECTORS

| Term | Definition |
|---|---|
| Norm/Length | $\|v\| = \sqrt{\sum_{i=1}^n v_i^2} = \sqrt{v^\top v}$ |
| | $\nabla\|v\| = \frac{1}{\|v\|}v^\top$ |
| | $\nabla\frac{1}{\|v\|} = -\frac{1}{\|v\|^2}\nabla\|v\| = -\frac{1}{\|v\|^3}v^\top$ |
| | $\nabla\|r\|^2 = 2\|r\|\frac{1}{\|r\|}r^\top = 2r^\top$ |
| Distance | $d(u,v) = \|u - v\|$ |
| | $\nabla d(u,v) = \frac{1}{d(u,v)}((u-v)^\top(v-u)^\top)$ |
| Scalar product | $u \cdot v = u^\top v = \sum_k u_k v_k$ |
| | $\nabla(u \cdot v) = (v^\top u^\top)$ |
| | $\nabla_u(u \cdot v) = v^\top$ |

## MATRICES

**Inverting**:

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \Rightarrow A^{-1} = \frac{1}{ad - cb}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

**Transposing**:

$$\begin{pmatrix} 1 & 0 \\ 0 & 2 \\ 3 & 1 \end{pmatrix}^\top = \begin{pmatrix} 1 & 0 & 3 \\ 0 & 2 & 1 \end{pmatrix} \quad (A^\top)^\top = A \quad (A+B)^\top = A^\top + B^\top$$

$$\begin{pmatrix} 1 \\ 4 \\ 5 \end{pmatrix}^\top = (1\ \ 4\ \ 5) \quad (\lambda A)^\top = \lambda A^\top \quad (AB)^\top = B^\top A^\top$$

**Multiplication**:

$$\begin{pmatrix} 2 & 3 & 1 \\ 4 & -1 & 7 \end{pmatrix} \cdot \begin{pmatrix} 6 & 4 \\ 1 & 0 \\ 8 & 9 \end{pmatrix}$$
$$= \begin{pmatrix} (2\cdot 6 + 3\cdot 1 + 1\cdot 8) & (2\cdot 4 + 3\cdot 0 + 1\cdot 9) \\ (4\cdot 6 + -1\cdot 1 + 7\cdot 8) & (4\cdot 4 + -1\cdot 0 + 7\cdot 9) \end{pmatrix} = \begin{pmatrix} 23 & 17 \\ 79 & 79 \end{pmatrix}$$

## PROPERTIES

$$A \cdot A^{-1} = \mathbb{1} \qquad \ker A = \{v \mid Av = 0\}$$
$$A + B = B + A \qquad A \cdot B \neq B \cdot A$$
$$A + (B + C) = (A + B) + C \qquad A(BC) = (AB)C$$
$$A(B + C) = AB + AC \quad (A + B)C = AC + BC$$

## DETERMINANT

$$\det(a) = a \qquad \det\begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - cb$$

$$\det\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32}$$
$$-a_{31}a_{22}a_{13} - a_{32}a_{23}a_{11} - a_{33}a_{21}a_{12}$$

$$\det(A \cdot B) = \det(A) \cdot \det(B) \quad \det(\mathbb{1}) = 1$$
$$\det(A^{-1}) = \frac{1}{\det(A)} \qquad \det(A^\top) = \det(A)$$

$$\det\begin{pmatrix} 0 & 2 & 1 \\ 1 & 0 & 1 \\ 3 & 4 & 2 \end{pmatrix} = -2\det\begin{pmatrix} 1 & 1 \\ 3 & 2 \end{pmatrix} + 0\det\begin{pmatrix} 0 & 1 \\ 3 & 2 \end{pmatrix} - 4\det\begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$$
$$= -2(2-3) + 0 - 4(-1) = 2 + 4 = 6$$

Schema : $\begin{pmatrix} + & - & + & \dots \\ - & + & - & \dots \\ + & - & + & \dots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}$

## EIGENVALUES

$$M \in \mathbb{R}^{n \times n}, \quad M^\top = M$$
$$\det(M) = \lambda_1 \cdot \lambda_2 \cdot \ldots \cdot \lambda_n$$

**Eigenvector $v$ for the eigenvalue $\lambda$ of $M$**:

$$Mv = \lambda v$$
$$\Leftrightarrow (M - \lambda\mathbb{1})v = 0$$

**Eigenvalue $\lambda$ of $M$**:

$$\det(M - \lambda\mathbb{1}) = 0$$

**Finding eigenvalues**:

$$A = \begin{pmatrix} 1 & 1 \\ -2 & 4 \end{pmatrix}$$
$$\det(A - \lambda\mathbb{1}) = \det\begin{pmatrix} 1-\lambda & 1 \\ -2 & 4-\lambda \end{pmatrix}$$
$$= (1-\lambda)(4-\lambda) - (-2)\cdot 1$$
$$= \lambda^2 - 5\lambda + 6$$
$$\Rightarrow \lambda \in \{3, 2\}$$

or alternatively

$$m = \frac{1}{2}\operatorname{tr}\begin{pmatrix} 1 & 1 \\ -2 & 4 \end{pmatrix} = \frac{1+4}{2} = \frac{\lambda_1 + \lambda_2}{2}$$
$$p = \det\begin{pmatrix} 1 & 1 \\ -2 & 4 \end{pmatrix} = 4 + 2 = \lambda_1\lambda_2$$
$$\lambda_{1,2} = m \pm \sqrt{m^2 - p} = \frac{5}{2} \pm \sqrt{\frac{25}{4} - 6}$$
$$= \frac{5}{2} \pm \frac{1}{2}$$

## INDICES

$$A \in \mathbb{R}^{n \times m}, B \in \mathbb{R}^{m \times r}, \qquad A \cdot B \in \mathbb{R}^{n \times r}$$
$$(A \cdot B)_{ij} = \sum_{k=1}^{m} A_{ik}B_{kj} = \sum_k A_{ik}B_{kj}$$
$$\left((A \cdot B)^\top\right)_{ij} = (A \cdot B)_{ji} = \sum_k A_{jk}B_{ki}$$
$$(A \cdot B^\top)_{ij} = \sum_k A_{ik}B^\top_{kj} = \sum_k A_{ik}B_{jk}$$
$$a^\top \cdot B \cdot b = \sum_{ij}(a^\top)_i B_{ij} b_j = \sum_{ij} a_i B_{ij} b_j$$
$$(B \cdot a)(C \cdot a) = \sum_i (B \cdot a)_i (C \cdot a)_i$$

### ADDITION/SUBTRACTION

$$A \in \mathbb{R}^{n \times m}, B \in \mathbb{R}^{n \times m}$$
$$(A + B)_{ij} = A_{ij} + B_{ij}$$
$$(A - B)_{ij} = A_{ij} - B_{ij}$$

## KRONECKER DELTA

$$\delta_{ij} = (\vec{e}_i)_j = \begin{cases} 1 & , i = j \\ 0 & , \text{otherwise} \end{cases}$$
$$(E_{ij})_{kr} = \delta_{ik}\delta_{jr}$$
$$(E_{rst})_{ijk} = \delta_{ri}\delta_{sj}\delta_{tk}$$

## BASIS VECTORS ($\mathbb{R}^3$)

$$e_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, e_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, e_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

## UNIT MATRICES ($\mathbb{R}^2$)

$$E_{11} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, E_{12} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, E_{21} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, E_{22} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

## AFFINE TRANSFORMATIONS

$$M \in \mathbb{R}^{m \times n}$$
$$A_{b,M} : \begin{cases} \mathbb{R}^n \to \mathbb{R}^n \\ x \mapsto b + M \cdot x \end{cases}$$
$$L_A(\vec{0}) = \vec{0}$$

## DERIVATIVES

| | | | |
|---|---|---|---|
| $1$ | $\to 0$ | $x$ | $\to 1$ |
| $x^2$ | $\to 2x$ | $x^n$ | $\to nx^{n-1}$ |
| $\frac{1}{x}$ | $\to -\frac{1}{x^2}$ | $\sqrt{x}$ | $\to \frac{1}{2\sqrt{x}}$ |
| $e^x$ | $\to e^x$ | $e^{-x}$ | $\to -e^{-x}$ |
| $\ln(x)$ | $\to \frac{1}{x}, x > 0$ | $\ln(y \cdot x)$ | $\to \frac{1}{x}, x > 0$ |
| $a^x$ | $\to \ln(a) \cdot a^x, a > 0, a \neq 1$ | $\log_b(x)$ | $\to \frac{1}{\ln(b) \cdot x}$ |
| $\sin(x)$ | $\to \cos(x)$ | $\sin(2x)$ | $\to 2\cos(2x)$ |
| $\cos(x)$ | $\to -\sin(x)$ | $\cos(ax)$ | $\to -a\sin(ax)$ |
| $\tan(x)$ | $\to 1 + \tan^2(x) = \frac{1}{\cos^2(x)}$ | $\operatorname{arccot}(x)$ | $\to -\frac{1}{1+x^2}$ |
| $\cot(x)$ | $\to -\frac{1}{\sin^2(x)}$ | $\arcsin(x)$ | $\to \frac{1}{\sqrt{1-x^2}}$ |
| $\arccos(x)$ | $\to -\frac{1}{\sqrt{1-x^2}}$ | $\arctan(x)$ | $\to \frac{1}{1+x^2}$ |

| Case | Rule |
|---|---|
| Addition | $(f(x) + g(x))' = f'(x) + g'(x)$ |
| Multiplication | $(a \cdot f(x))' = a \cdot f'(x)$ |
| Product rule | $(f \cdot g)' = f' \cdot g + f \cdot g'$ |
| | $(f \cdot g \cdot h)' = f' \cdot g \cdot h + f \cdot g' \cdot h + f \cdot g \cdot h'$ |
| Chain rule | $(f(g(x)))' = f'(g(x)) \cdot g'(x)$ |
| Quotient rule | $\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$ |

## INTEGRALS

| | | | |
|---|---|---|---|
| $\frac{1}{x}$ | $\to \ln\lvert x\rvert + c$ | $\ln(x)$ | $\to x\ln(x) - x + c$ |
| $e^x$ | $\to e^x + c$ | $\frac{1}{\sqrt{1-x^2}}$ | $\to \arcsin(x) + c$ |
| $\sin(x)$ | $\to -\cos(x) + c$ | $-\frac{1}{\sqrt{1-x^2}}$ | $\to \arccos(x) + c$ |
| $\cos(x)$ | $\to \sin(x) + c$ | $\frac{1}{1+x^2}$ | $\to \arctan(x) + c$ |
| $\tan(x)$ | $\to -\ln\lvert\cos(x)\rvert) + c$ | $a^{b \cdot x}$ | $\to \frac{1}{\ln(a^b)}a^x + c$ |
| $x^a$ | $\to \frac{1}{a+1}x^{a+1} + c, a \neq -1$ | | |
| $a^x$ | $\to \frac{1}{\ln(a)}a^x + c$ | | |

$a^x = e^{\ln(a) \cdot x}$

$$\arctan(x) \to x\arctan(x) - \frac{1}{2}\ln(|1 + x^2|) + c$$
$$\arccos(x) \to x\arccos(x) - \sqrt{1-x^2} + c$$
$$\arcsin(x) \to x\arcsin(x) + \sqrt{1-x^2} + c$$

## RULES

$$\int f(g(x))g'(x)\,dx = F(g(x)) + \text{const}$$
$$\int f(x)g(x)\,dx = F(x)g(x) - \int F(x)g'(x)\,dx$$

## FUNCTIONS OF SEVERAL VARIABLES

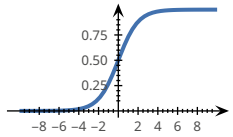
| Term | Definition |
|---|---|
| Real valued function | $R \subset \mathbb{R}$ |
| Vector valued function | $R \subset \mathbb{R}^m$ |
| A function of $n$ variables | $D \subset \mathbb{R}^n$ |
| $n$-dimensional curve | $c : \mathbb{R} \to \mathbb{R}^n$ |
| $n$-dimensional hyper-surface | $s : \mathbb{R}^n \to \mathbb{R}$ |
| Reparametrization | $c(t) \to d(s) = c(h(s))$ |

**Softmax function**:

$$\sigma : \begin{cases} \mathbb{R}^n \to [0;1]^n \\ x_i \mapsto e^{x_i}\left(\sum_{j=1}^n e^{x_j}\right)^{-1} \end{cases}$$
$$J_{ij}\sigma(x) = \delta_{ij}\frac{e^{x_i}}{\sum_{k=1}^n e^{x_k}} - \frac{e^{x_i}e^{x_j}}{\left(\sum_{k=1}^n e^{x_k}\right)^2}$$

**Sigmoid function**:

$$\sigma : \begin{cases} \mathbb{R} \to [0;1] \\ x \mapsto \frac{1}{1+e^{-x}} = \frac{e^x}{1+e^x} \end{cases}$$
$$\sigma(-x) = 1 - \sigma(x)$$
$$\sigma'(x) = \sigma(x)(1 - \sigma(x))$$




## DERIVATIVES

| Function | Derivative |
|---|---|
| $f : \begin{cases} \mathbb{R} \to \mathbb{R}^n \\ x \mapsto y \end{cases}$ | $f' : \begin{cases} \mathbb{R} \to \mathbb{R}^n \\ x \mapsto y \end{cases} = \begin{pmatrix} \frac{d}{dx}f_1(x) \\ \frac{d}{dx}f_2(x) \\ \vdots \\ \frac{d}{dx}f_n(x) \end{pmatrix}$ |
| $f : \begin{cases} \mathbb{R}^n \to \mathbb{R} \\ x \mapsto y \end{cases}$ | $f' : \begin{cases} \mathbb{R}^n \to \mathbb{R}^{1\times n} \\ x \mapsto y \end{cases} = \nabla f$ <br> $= \left(\frac{\partial}{\partial x_1}f(x), \frac{\partial}{\partial x_2}f(x), \ldots, \frac{\partial}{\partial x_n}f(x)\right)$ |
| $f : \begin{cases} \mathbb{R}^n \to \mathbb{R}^m \\ x \mapsto y \end{cases}$ | $f' : \begin{cases} \mathbb{R}^n \to \mathbb{R}^{m\times n} \\ x \mapsto y \end{cases} = J_f = \left.\frac{\partial f}{\partial x}\right\rvert_{x=x_0}$ <br> $= \begin{pmatrix} \nabla f_1(x) \\ \vdots \\ \nabla f_m(x) \end{pmatrix} = \begin{pmatrix} \frac{\partial}{\partial x_1}f_1(x) & \ldots & \frac{\partial}{\partial x_n}f_1(x) \\ \vdots & \ddots & \vdots \\ \frac{\partial}{\partial x_1}f_m(x) & \ldots & \frac{\partial}{\partial x_n}f_m(x) \end{pmatrix}$ |

### PARTIAL VS TOTAL DERIVATIVE

$$f(x, y) = x^2 + 2y$$
$$\frac{\partial f}{\partial x} = 2x$$
$$\frac{df}{dx} = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} \cdot \frac{dy}{dx} = 2x + 2 \cdot \frac{dy}{dx} \mid \text{(y dependent on x)}$$

Let $y = x^3 \Rightarrow \frac{df}{dx} = 2x + 6x^2$

### GRADIENT VS TOTAL DERIVATIVE

$$f(x, y) = xye^y$$
$$x(t) = e^t$$
$$y(t) = t^2$$
$$\nabla f = (ye^y, xe^y + xye^y) \qquad = e^y(y, x + xy)$$
$$\frac{d}{dt}f = e^y(y, x + xy) \cdot \frac{d}{dt}\begin{pmatrix} e^t \\ t^2 \end{pmatrix} = \frac{d}{dt}\left(e^t t^2 e^{t^2}\right)$$
$$= \left(2t + t^2 + 2t^3\right)e^{t^2+t}$$

## GRADIENTS

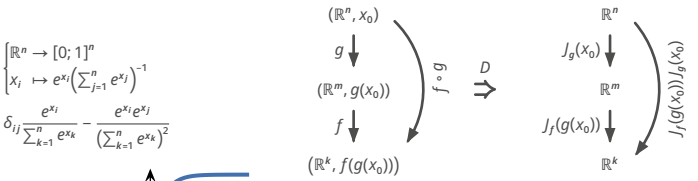
**Common gradients**:

$$f : \mathbb{R}^n \to \mathbb{R},\ x \in \mathbb{R}^n,\ a \in \mathbb{R}^n,\ c \in \mathbb{R},\ A \in \mathbb{R}^{n \times n}$$
$$f(x) = \|x\|^2 \qquad \Rightarrow \nabla f = 2x^\top$$
$$f(x) = a^\top x + c \qquad \Rightarrow \nabla f = a^\top$$
$$f(x) = x^\top Ax \qquad \Rightarrow \nabla f = x^\top(A + A^\top)$$
$$f(x) = \frac{1}{2}x^\top Ax + b^\top x + c \quad \Rightarrow \nabla f = x^\top\frac{A + A^\top}{2} + b^\top$$
$$= x^\top A + b^\top \text{ if } A^\top = A$$
$$g : \mathbb{R}^m \to \mathbb{R}, B \in \mathbb{R}^{m \times n}$$
$$f(x) = g(Bx + b) \qquad \Rightarrow \nabla f = (\nabla g(Bx + b))^\top B$$

## JACOBIAN MATRIX

$$f : \mathbb{R}^n \to \mathbb{R}^m \qquad x_0 \in \mathbb{R}^n$$
$$J_f(x_0) : \mathbb{R}^{m \times n} \qquad J_f(x_0)_{ij} = \frac{\partial}{\partial x_j}f_i(x_0)$$
$$f : \mathbb{R}^2 \to \mathbb{R}^2 \quad J_f(x, y) = \begin{pmatrix} \partial x_1 & \partial y_1 \\ \partial x_2 & \partial y_2 \end{pmatrix}$$

Euclid$_*$ ty lukas <3 Mat$_{\mathbb{R}}$




**Chain rule**:

$$f : \mathbb{R}^n \to \mathbb{R}^m,\ g : \mathbb{R}^m \to \mathbb{R}^k,\ h : \mathbb{R}^n \to \mathbb{R}^k = g \circ f$$
$$J_h = J_g(f(x)) \cdot J_f(x)$$

**Common jacobians**:

$$f(x) = a + M \cdot x \qquad \Rightarrow J_f = M$$
$$f(x) = x \qquad \Rightarrow J_f = \mathbb{1}$$
$$f(x) = (x^\top x)x \qquad \Rightarrow J_f = 2xx^\top + (x^\top x) \cdot \mathbb{1}$$
$$\Rightarrow (J_f)_{ij} = 2x_j x_i + \sum_k x_k^2\delta_{ij}$$
$$f(x) = \|x\|^2 \qquad \Rightarrow J_f = (2x)^\top$$
$$f_i : \begin{cases} \mathbb{R}^n \to \mathbb{R}^n \\ x \mapsto a_i + M_i x \end{cases}$$
$$g(x) = f_n(f_{n-1}(\ldots(f_1(x)))) \Rightarrow J_g = M_n \cdot M_{n-1} \cdot \ldots \cdot M_1$$

### LINEARISATION

$$f : \mathbb{R}^n \to \mathbb{R}^m,\ x_0 \in \mathbb{R}^n$$
$$L_f(x) = f(x_0) + J_f(x_0) \cdot (x - x_0)$$

For finding better approximations for $x_0$:
1) Calculate $J_f$, then $J_f(x_0)$, then $L_f$
2) Find $x$ so that $L_f(x) = 0$ (Gauss)

For calculating an approximate value at $x$:
1) $x_0$ = round $x$ to ints
2) Calculate $L_f$ at $x_0$
3) Calculate $L_f(x)$ with original $x$

## SECOND DERIVATIVE

$$\frac{\partial}{\partial x}\left(\frac{\partial}{\partial x}(f(x, y))\right) = \frac{\partial^2 f(x, y)}{\partial x^2} = \partial_{xx}f(x, y)$$

## HESSIAN MATRIX

$$f : \mathbb{R}^n \to \mathbb{R}, \quad x \in \mathbb{R}^n, \quad H(x) \in \mathbb{R}^{n \times n}$$
$$H(x, y) = \begin{pmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{pmatrix} = J(\nabla f)$$
$$(H(x))_{ij} = \frac{\partial}{\partial x_i}\frac{\partial}{\partial x_j}f(x)$$
$$\partial_i\partial_j f = \partial_j\partial_i f$$

**Common hessians**:

$$f(x) = \frac{1}{2}x^\top Ax + b^\top x + c \Rightarrow H_f = \frac{A + A^\top}{2}$$
$$f(x) = g(Ax + b) \qquad \Rightarrow H_f = A^\top H_g(Ax + b)A$$
$$g : \mathbb{R}^m \to \mathbb{R}$$

## EXTREMAL POINTS

**Quadratic form**:

$$M \in \mathbb{R}^{n\times n} \qquad q_M(v) = v^\top M v \qquad v \in \mathbb{R}^n$$

$$H = \frac{1}{2}(M + M^\top) \quad \forall v \in \mathbb{R}^n.\ (q_M(v) = q_H(v))$$

$\forall v.\ (v^\top H v > 0) \vee \forall \lambda.\ (\lambda > 0) \Rightarrow c_0$ is a ***local minimum***, $H$ is ***positive definite***

$\forall v.\ (v^\top H v \geq 0) \vee \forall \lambda.\ (\lambda \geq 0) \Rightarrow c_0$ is a ***local minimum*** or ***non-extremal***, $H$ is ***positive semi-definite***

$\forall v.\ (v^\top H v < 0) \vee \forall \lambda.\ (\lambda < 0) \Rightarrow c_0$ is a ***local maximum***, $H$ is ***negative definite***

$\forall v.\ (v^\top H v \leq 0) \vee \forall \lambda.\ (\lambda \leq 0) \Rightarrow c_0$ is a ***local maximum*** or ***non-extremal***, $H$ is ***negative semi-definite***

$\forall v.\ (v^\top H v \neq 0) \vee \forall \lambda.\ (\lambda \neq 0) \Rightarrow c_0$ is a ***saddle point***, $H$ is ***in-definite***

**Finding stationary points**:

1) Calculate $\nabla f$, then $H_f$
2) Find stationary point(s) $x$ so that $\nabla f(x) \overset{!}{=} 0$
3) Find type by technique of "completing squares" on $H_f(x)$ or finding its Eigenvalues

$$(v_1, v_2, v_3) \cdot \begin{pmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{pmatrix} \cdot \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}$$

$$= v_1^2 h_{11} + v_2^2 h_{22} + v_3^2 h_{33} + 2v_1 v_2 h_{12} + 2v_2 v_3 h_{23} + 2v_1 v_3 h_{13}$$

**Technique of completing squares**:

$$\underbrace{x^2}_{\sqrt{\cdot}} \underbrace{- 4xy}_{\cdot 2x} + 1$$

$$= (x - 2y)^2 - (-2y)^2 + 1$$

$$= (x - 2y)^2 - 4y^2 + 1$$

## HYPER-SURFACES

**Chain rule**:

$$f : \mathbb{R}^n \to \mathbb{R}^m,\ g : \mathbb{R}^m \to \mathbb{R},\ h : \mathbb{R}^n \to \mathbb{R} = g \circ f$$

$$\nabla h(x) = \nabla g(f(x)) = \nabla g(f)|_{f=f(x)} \cdot \frac{\partial}{\partial x} f(x)$$

**Differentiation properties**:

$$f : \mathbb{R}^n \to \mathbb{R},\ g : \mathbb{R}^n \to \mathbb{R}$$

$$\nabla(f + g) = \nabla f + \nabla g$$

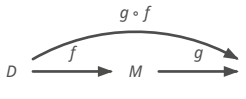
$$\nabla(f - g) = \nabla f - \nabla g$$

$$\nabla(f \cdot g) = g\nabla f + f\nabla g$$

$$\nabla \frac{f}{g} = \frac{g\nabla f - f\nabla g}{g^2}$$

## COMMUTATIVE DIAGRAMS

$f : D \to M$
$g : N \to R$
$h : D \to R$
$= g \circ f$




## GRADIENT DESCENT / NEWTON'S METHOD

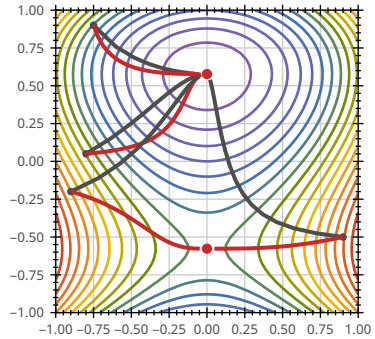
$$GD \quad x_{i+1} = x_i - \gamma \cdot \nabla f(x_i)$$

$$N \quad x_{i+1} = x_i - \gamma \cdot \left(H_{f(x_i)}\right)^{-1} \cdot \nabla f(x_i)$$

$\gamma$ = step size

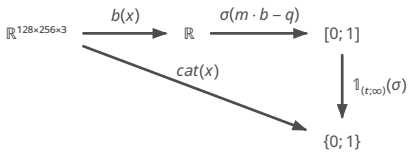
termination criterion $\varepsilon > |x_{i+1} - x_i|$

Gradient descent doesn't always produce global minimum:
– If the function has more than one local minimum it could output that → run with different starting points
– Could output a stationary point that is not minimal (unlikely). → run with different starting points
– Could not terminate at all if step size / learning rate is so big (overshoots minimal point). Likely to happen if the graph of the function is narrow around the minimal point. → choose smaller step size




## LOGISTIC REGRESSION

| *Term* | *Definition* |
|---|---|
| Probability measure | $\mathbb{P}(E)$ |
| Density | $f_\theta(x_i)$ |
| Likelihood function | $L(\theta) = \prod_{i=1}^{n} f_\theta(x_i)$<br>e.g. $L(\theta) = \prod_{x\in S_{Yes}} p_\theta(f(x)) \prod_{x\in S_{No}} 1 - p_\theta(f(x))$ |
| Log-likelihood function | $\ln(L(\theta)) = \sum_{i=1}^{n} \ln(f_\theta(x_i))$ |
| Cost function (negative log-likelihood) | $c(\theta) = -\ln(L(\theta)) = -\sum_{i=1}^{n} \ln(f_\theta(x_i))$ |
| Indicator function | $\mathbb{1}_{(a;b)}(x) = \begin{cases} 1 & x \in (a;b) \\ 0 & x \notin (a;b) \end{cases}$ |




$b(x)$ = Feature function
$\sigma(m \cdot b - q)$ = Probability function
$m$ = Weight
$q$ = Bias
$\mathbb{1}_{(t;\infty)}(\sigma)$ = Indicator function
$t$ = Threshold

## MAXIMUM LIKELIHOOD PRINCIPLE

= Finding optimal parameters $\theta$ for fitting distribution to data

1) Calculate cost function $c$ with given distribution function $f_\theta$, parameters $\theta$ and data points $x_1, x_2, ..., x_n$
2) Calculate stationary point(s) of $c$ (set $\nabla c$ to 0)
   ⇒ Find optimum (gradient descent/newton's method if calculation not possible)

## PROBABILITY THEORY

| *Term* | *Definition* |
|---|---|
| Distributed | ~ |
| Normal distribution | $\mathcal{N}(\mu, \sigma)$ |
| Uniform distribution | unif$(a, b)$ |
| *CDF* | Cumulative Distribution Function $F$ |
| *PDF* | Probability Density Function $f$ |

## KOLMOGOROV AXIOMS

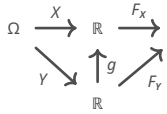
$$\mathbb{P}(\Omega) = 1$$

$$\mathbb{P}(\varnothing) = 0$$

$$\forall A, B \subset \Omega,\ A \cap B = \varnothing.\ (\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B))$$

$$\forall A, B \subset \Omega.\ (\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B))$$

## RELATING RANDOM VARIABLES




$$g(Y(\omega)) = X(\omega)$$

$$\Rightarrow F_X(g(y)) = P(X(\omega) \leq g(y)) = P(g(Y(\omega)) \leq g(y))$$

$$\Rightarrow \begin{cases} F_X(g(y)) = P(Y(\omega) \leq y) = F_Y(y) & g \text{ increasing} \\ F_X(g(y)) = P(Y(\omega) \geq y) = 1 - F_Y(y) & g \text{ decreasing} \end{cases}$$

$$\Rightarrow \begin{cases} f_X(g(y)) \cdot g'(y) = f_Y(y) & g \text{ increasing} \\ f_X(g(y)) \cdot g'(y) = -f_Y(y) & g \text{ decreasing} \end{cases}$$

$$|f_Y(y)\,\mathrm{d}y| = |f_X(x)\,\mathrm{d}x| \quad \text{if monotonically in/decreasing}$$

The bounds $a, b$ of the indicator function $\mathbb{1}$ inside $f_Y(y)$ must be swapped relative to $f_X(x)$ if $g$ is monotonically decreasing!

$$Y(\omega) = h(X(\omega)) \quad \mathbb{1}_{(a;b)} \to \mathbb{1}_{(h(b);h(a))}$$

## DISCRETE PROBABILITY DISTRIBUTION

Let $\Omega = \{e_1, e_2, ..., e_n\}$ be enumerable

$$\mathbb{P}(\{e_k\}) = p_k, \quad k \in \{1, 2, ..., n\}, \quad E \subset \Omega$$

$$\mathbb{P}(E) = \sum_{e\in E} p(e)$$

$$F(x) = \sum_{e\in\Omega, e\leq x} p(e)$$

### RULES

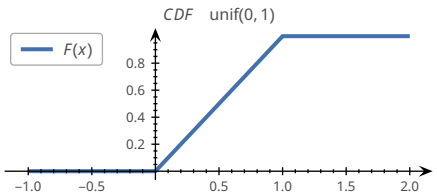
$$\forall e \in \Omega.\ (0 \leq p(e) \leq 1)$$

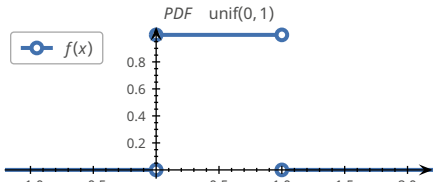
$$\sum_{e\in\Omega} p(e) = 1 \qquad \int_{-\infty}^{\infty} f(t)\,\mathrm{d}t = 1$$

$$\lim_{x\to-\infty} F(x) = 0 \qquad \lim_{x\to\infty} F(x) = 1$$

$$0 \leq F(x) \leq 1 \qquad f(t) \geq 0 \text{ for } t \in \mathbb{R}$$

$F(x)$ is monotonically increasing & continuous from the right




## UNIVARIATE UNIFORM DISTRIBUTION

$X \sim \text{unif}(a, b), \quad \Omega \subset \mathbb{R}$

$$f_X(x) = \begin{cases} \frac{1}{b-a} & \text{if } x \in (a;b) \\ 0 & \text{else} \end{cases} = \mathbb{1}_{(a;b)}(x) \cdot \frac{1}{b-a} = F_X'$$

$$F_X(x) = \mathbb{P}((-\infty; x] \cap \Omega) = \mathbb{1}_{[a;b)}(x) \cdot \frac{x-a}{b-a} + \mathbb{1}_{(b;\infty)}(x)$$

$$= \mathbb{P}(X \leq x) = \int_{-\infty}^{x} f_X(t)\,\mathrm{d}t$$

## UNIVARIATE NORMAL DISTRIBUTION

**Standard normal distribution**:

$$\varphi(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$$

**General normal distribution**:

$$f(x|\mu, \sigma) = \frac{1}{\sigma}\varphi\left(\frac{x-\mu}{\sigma}\right)$$

**Univariate normal distribution**:

$$x \sim \mathcal{N}(\mu, \sigma)$$

$$f(x|\mu, \sigma) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}$$

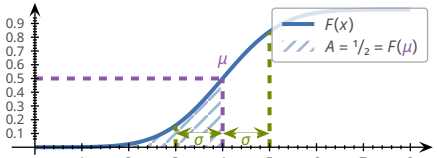
$$\mu \approx \bar{x} = \frac{1}{N}\sum_{i=1}^{N} x_i$$

$$\sigma^2 \approx \text{var} = \frac{1}{N-1}\sum_{i=1}^{N} (x_i - \bar{x})^2$$

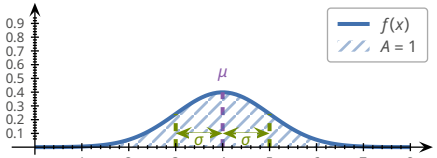
**Error function**:

$$\text{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2}\,\mathrm{d}t$$

*CDF*




*PDF*




mean value ($\mu$), standard deviation ($\sigma$)

## MULTIVARIATE NORMAL DISTRIBUTION

$\Sigma$ = covariance matrix, $V : \Omega \to \mathbb{R}^n$

$$V \sim \mathcal{N}(\mu, \Sigma)$$

$$f_V(v) = \frac{1}{\sqrt{(2\pi)^n \det\Sigma}} \cdot e^{q_{\Sigma^{-1}}(v-\mu)}$$

$$q_{\Sigma^{-1}}(v - \mu) = -\frac{1}{2}(v-\mu)^\top \Sigma^{-1}(v-\mu)$$

**Sample mean**:

$$\mu \approx \bar{v} = \frac{1}{N}\sum_{i=1}^{N} v_i$$

**Sample covariance matrix**:

$$\Sigma \approx Q = \frac{1}{N-1}\sum_{i=1}^{N} (v_i - \bar{v})(v_i - \bar{v})^\top$$

### STATISTICALLY INDEPENDENT

$X : \Omega \to \mathbb{R}$ and $Y : \Omega \to \mathbb{R}$ are ***statistically independent***, if the probability density $f_V$ of the random variable

$$V : \begin{cases} \Omega \to \mathbb{R}^2 \\ \omega \mapsto \begin{pmatrix} X(\omega) \\ Y(\omega) \end{pmatrix} \end{cases}$$

satisfies

$$f_V(x, y) = f_X(x) \cdot f_Y(y)$$

## EXAMPLES

## WORKING WITH INDICES

$x, y \in \mathbb{R}^n,\ f : \mathbb{R}^n \to \mathbb{R}$

$$f(x) = (x - y)^\top (3x - y) = \sum_{i=1}^{n} (x_i - y_i)(3x_i - y_i)$$

$$\partial x_i f(x) = (3x_i - y_i) + 3(x_i - y_i)$$

$$f(x) = \ln(y^\top x) = \ln\left(\sum_{i=1}^{n} y_i x_i\right) \Rightarrow \nabla f = \frac{1}{y^\top x} y^\top$$

$$f : \begin{cases} \mathbb{R}^{n\times n} \to \mathbb{R}^{n\times n} \\ X \mapsto X \cdot A \end{cases} \qquad \frac{\partial}{\partial x_{ij}} f(X) \overset{!}{=} E_{ij} \cdot A$$

$$\frac{\partial}{\partial x_{ij}} f_{kl}(X) = \frac{\partial}{\partial x_{ij}} \sum_r x_{kr} a_{rl} = \sum_r \delta_{ik}\delta_{jr} a_{rl} = (E_{ij} \cdot A)_{kl}$$

## CALCULATING LIKELIHOOD FUNCTION

Given the following data and the probability density of the form

$$f(t) = \lambda e^{-\lambda t} \cdot 1_{(0;\infty)}(t)$$

find a formula for the likelihood function:

| $x$ | $y$ |
|---|---|
| 1 | 7 |
| 2 | 16 |
| 3 | 32 |
| 4 | 44 |

$$l(\lambda) = f(7) \cdot f(16) \cdot f(32) \cdot f(44) = \lambda^4 e^{-99\lambda}$$

Find a formula for the negative log-likelihood:

$$-\ln(l(\lambda)) = -\ln\left(\lambda^4 e^{-99\lambda}\right) = 99\lambda - 4\ln(\lambda)$$

## CALCULATING PROBABILITY DENSITY FUNCTION

### EXPONENTIAL DISTRIBUTION

Given $\sigma$, $Y$ and $X$, an exponentially distributed random var

$$f_X(x) = X_{[0;\infty)}(x) \cdot e^{-x}, \quad Y = \sigma(X), \quad \sigma(x) = \frac{1}{1 + e^{-x}}$$

calculate the probability density function $f_Y(y)$

$$|f_Y(y)\,\mathrm{d}y| = |f_X(x)\,\mathrm{d}x|$$

$$\overset{\frac{\mathrm{d}x}{\mathrm{d}y}>0}{\Leftrightarrow} f_Y(y) = f_X(x) \cdot \frac{\mathrm{d}x}{\mathrm{d}y}$$

$$y = \frac{1}{1 + e^{-x}}$$

$$\Leftrightarrow x = -\ln\left(\frac{1}{y} - 1\right), \quad 0 < y < 1$$

$$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}y} = -\frac{1}{\frac{1}{y} - 1} \cdot \frac{-1}{y^2} = \frac{1}{y - y^2}$$

$$\Rightarrow f_Y(y) = X_{[0;\infty)}(x) \cdot e^{-x} \cdot \frac{1}{y - y^2}$$

$$= \frac{1}{y^2} \cdot X_{[0;\infty)}\left(-\ln\left(\frac{1}{y} - 1\right)\right)$$

$$-\ln\left(\frac{1}{y} - 1\right) \geq 0 \overset{y>0}{\Leftrightarrow} y \geq \frac{1}{2}$$

$$\Rightarrow X_{[0;\infty)}\left(-\ln\left(\frac{1}{y} - 1\right)\right) \overset{y\in(0;1)}{=} X_{[\frac{1}{2};1)}(y)$$

$$\Rightarrow f_Y(y) = \frac{1}{y^2} X_{[\frac{1}{2};1)}(y)$$

### UNIFORM DISTRIBUTION

Given $X \sim \text{unif}(0, 1)$, $p > 0$, $Y = X^p$ calculate the probability density function $f_Y(y)$

$$F_X(x) = \mathbb{P}(X \leq x)$$

$$F_Y(y) = \mathbb{P}(Y \leq y) = \mathbb{P}(X^p \leq y)$$

$$= \mathbb{P}\left(X \leq y^{\frac{1}{p}}\right) = F_X\left(y^{\frac{1}{p}}\right)$$

$$= \mathbb{1}_{[0;1]}(y) y^{\frac{1}{p}} + \mathbb{1}_{1;\infty)}(y)$$

$$f_Y(y) = \frac{\mathrm{d}}{\mathrm{d}y} F_Y(y) = \mathbb{1}_{[0;1]}(y) \frac{1}{p} y^{\frac{1}{p}-1}$$

### UNIFORM DISTRIBUTION 2

$$X \sim \text{unif}(1, 3), \quad Y = \sqrt[3]{\frac{5 - X}{2}}$$

$$f_X(x) = \mathbb{1}_{(1;3)}(x)\frac{1}{2}$$

$$X = 5 - 2Y^3$$

$$g(y) = 5 - 2y^3$$

$$f_Y(y) = -f_X(g(y))g'(y) \qquad \text{monot. decr.}$$

$$= 3y^2 \mathbb{1}_{(1;3)}(5 - 2y^2)$$

$$1 < 5 - 2y^2 < 3$$

$$\Leftrightarrow 1 < y < \sqrt[3]{2}$$

$$\Rightarrow f_Y(y) = 3y^2 \mathbb{1}_{(1;\sqrt[3]{2})}(y)$$